

Infosys Springboard Virtual Internship 7.0

Milestone 4 report

Batch Number : Group 1

Start Date : 29-06-2026

Team Details : A

Team Members :

Kapu Tejaswini

Challa Jaya Deepika Reddy

Internship Duration : 8 weeks

1. Project Title:

Supply Chain Visibility System with Optimization Analytics Group 1

2. Project Objective :

The main objective of this project is to **analyze and visualize supply chain data to understand overall business, product, customer, delivery, market, order, discount, profit, and shipping performance.**

The project aims to:

- Preprocess and clean the supply chain dataset by removing unnecessary data, handling errors, changing data types, and creating useful derived columns.
- Analyze **sales, orders, profit, products, customers, markets, and delivery performance.**
- Develop **7 interactive Power BI dashboards** to present different aspects of supply chain performance.
- Identify important trends and patterns in **product sales, pricing, customer distribution, shipping, regional performance, discounts, and profitability.**
- Provide meaningful insights that can support **better business and supply chain decision-making.**
- Present the analysis through clear and interactive visualizations for easier understanding of the supply chain operations.

3. Project Description in detail:

3.1 Overview

The **Supply Chain Visibility and Optimization** project focuses on analyzing supply chain data to understand different aspects of supply chain operations. The project uses information related to **orders, products, customers, sales, pricing, discounts, profits, markets, delivery status, shipping, and regions.** The data was first cleaned and transformed using **Power Query**, where unnecessary columns were removed, duplicates and errors were handled, data types were corrected, values were replaced where required, and useful columns such as **Year and Quantity Range** were created.

After preprocessing, the cleaned dataset was used to develop **seven interactive Power BI dashboards**: Supply Chain Performance, Product Performance & Pricing, Customer & Geographic Analysis, Delivery & Shipping Performance, Regional & Market Performance, Order & Discount Performance, and Profitability & Order Insights. These dashboards provide insights into **sales, orders, product performance, customer distribution, delivery efficiency, regional and market performance, discount patterns, and profitability**, helping to understand supply chain performance and support data-driven decision-making.

3.2 Project Approach

The project followed a structured and systematic approach to analyze the supply chain dataset and develop interactive Power BI dashboards. Each phase was carefully planned to ensure accurate data analysis and meaningful business insights. The overall project execution included data collection, preprocessing, visualization, dashboard development, and insight generation.

Step 1: Problem Statement

- Identified the need to improve supply chain visibility and analyze business performance using Power BI dashboards.

Step 2: Data Collection

- Collected the DataCo Supply Chain dataset and imported it into Microsoft Power BI.

Step 3: Data Understanding

- Studied the dataset structure, attributes, data types, and relationships between different columns.

Step 4: Data Preprocessing

- Performed preprocessing using Power Query Editor by correcting data types, splitting customer names, extracting the year, and transforming data into an analysis-ready format.

Step 5: Data Cleaning

- Removed unnecessary columns, handled missing values, removed duplicates, filtered invalid records, and ensured data quality.

Step 6: Data Visualization

- Created KPI cards, bar charts, line charts, pie charts, donut charts, treemaps, maps, scatter plots, tables, matrix, waterfall charts, ribbon charts, and slicers to analyze different business metrics.

Step 7: Dashboard Development

- Developed five interactive dashboards:
 - Supply Chain Performance Dashboard
 - Product Performance & Pricing Dashboard
 - Customer & Geographical Analysis Dashboard

- Delivery & Shipping Performance Dashboard
- Regional & Market Performance Dashboard
- Order & Discount Performance Dashboard
- Profitability & Order Insights Dashboard

Step 8: Insights & Analysis

- Analyzed sales, profit, product performance, customer behavior, shipping performance, delivery status, and regional performance to generate business insights for decision-making.

3.3 Tools and Technologies Used :

Tools	Purpose
DataCo Supply Chain Dataset (CSV)	Used as the primary dataset for supply chain analysis and imported into Power BI.
Microsoft Power BI Desktop	Used for data visualization, dashboard creation, and business intelligence reporting.
Power Query Editor	Used for data cleaning, transformation, preprocessing, and feature engineering.
DAX (Data Analysis Expressions)	Used to create calculated measures and KPIs such as Total Sales, Total Profit, Total Orders, and Average Shipping Days.
Microsoft Excel	Used to view and verify the CSV dataset before importing it into Power BI.

3.4 Insights from the Dashboard :

3.4a Supply Chain Performance Dashboard Insights :

- The company processed more than **180K orders**, indicating large-scale supply chain operations.
- **Europe** emerged as the highest sales-generating market.
- Late deliveries represent a significant portion of delivery statuses, indicating opportunities for logistics improvement.
- The average actual shipping time provides an overview of overall delivery efficiency.
- Year-wise sales trends help identify changes in overall business performance over time.

3.4b Product Performance & Pricing Dashboard Insights :

- The analysis identifies the **top-performing products** based on sales contribution.
- Product category comparison helps identify categories with higher sales performance.

- Department-wise analysis highlights the departments contributing more to overall sales.
- Product price and sales analysis helps understand the relationship between pricing and product sales.
- Product performance insights can support better **inventory planning and pricing strategies**.

3.4c Customer & Geographic Analysis Dashboard Insights :

- Customer distribution varies across different states and geographical locations.
- The **Consumer segment** represents a major share of customer activity.
- Average sales per customer helps evaluate customer-level sales contribution.
- State- and city-level analysis identifies locations with stronger sales performance.
- Customer and geographic insights can support **region-specific marketing and customer relationship strategies**.

3.4d Delivery & Shipping Performance Dashboard Insights :

- The analysis identifies a significant number of **late deliveries**, highlighting opportunities to improve logistics efficiency.
- Different shipping modes show differences in delivery-status distribution.
- Market-wise benefit comparison helps identify markets contributing more to profitability.
- Scheduled shipping days provide useful information for evaluating delivery planning.
- Year-wise shipping performance analysis helps monitor changes in logistics performance over time.

3.4e Regional & Market Performance Dashboard Insights :

- The analysis identifies differences in sales performance across the five major markets.
- **Europe** shows strong market performance compared with the other markets.
- Customer segment comparison helps identify the segments contributing more to sales and orders.
- Actual sales performance can be compared with the defined **sales target** to evaluate target achievement.
- Shipping-mode analysis shows how orders are distributed across different delivery methods.

3.4f Order & Discount Performance Dashboard Insights :

- The dashboard shows the overall **quantity of products sold**, providing an indication of order volume.
- Quantity-level analysis shows that a large portion of orders fall within the **1–2 quantity range**.
- Discount-rate analysis helps monitor the level of discounts applied to order items.
- Order-status distribution across markets helps identify differences in order processing across regions.
- The relationship between **discount rate and sales by product** provides insights into how discounting may be associated with product sales.
- The decomposition analysis allows total sales to be examined by **market, product category, and shipping mode**.

3.4g Profitability & Order Insights Dashboard :

- Average profit ratio provides an overview of the **profitability of order items**.
- Total discount amount helps understand the overall value of discounts provided to customers.
- Average benefit by shipping mode helps compare profitability associated with different shipping methods.
- Department-wise sales trends show how sales performance changes across departments over different years.
- The discount trend helps identify changes in discount levels over time.
- Overall order-status analysis provides insights into the distribution of orders across different processing and completion statuses.

3.5 Real-World Impact for Media and Public Communication :

The project demonstrates how **data visualization and business intelligence** can be used to convert complex supply chain information into clear and understandable insights. The Power BI dashboards help organizations communicate important information about **sales, products, customers, delivery performance, markets, orders, discounts, and profitability** in an interactive and easy-to-understand format.

The dashboards can support better communication between different departments by providing a **common view of business performance**. Managers and decision-makers can quickly identify delivery issues, understand regional and customer trends, monitor sales and profitability, and communicate important findings using visual reports. This improves **operational transparency, data-driven communication, strategic planning, and decision-making** across the organization.

The interactive nature of the dashboards allows users to **filter and explore information based on different business requirements**, making the analysis more flexible and useful.

Visual representations such as charts, maps, KPI cards, tables, and other interactive visuals make complex data easier to interpret and present to different audiences.

The project also shows how data-driven communication can help organizations **identify performance gaps, compare business regions, monitor operational activities, and recognize areas for improvement**. These insights can be effectively shared through reports and presentations, supporting clear communication between management, operational teams, and other stakeholders. Overall, the project demonstrates the practical value of **Power BI-based data visualization for transparent reporting and informed business decisions**.

4. Timeline Overview :

Week	Activities Planned	Activities Completed
Week1	• Collect the DataCo Supply Chain Dataset. • Understand dataset structure and attributes. • Perform data preprocessing and cleaning. • Transform data into analysis-ready format. • Create Power BI dashboard and generate insights.	• DataCo Supply Chain Dataset collected • Dataset attributes and important features understood • Data preprocessing performed using Power Query Editor • Unnecessary columns removed, duplicates handled, and data types corrected • Cleaned dataset connected to Power BI • KPI cards, charts, and slicers created for dashboard visualization
Week2	• Enhance Power BI dashboards with additional report pages • Create multiple dashboard visualizations for different business perspectives • Analyze dashboard results and generate business insights • Document dashboard development and project findings • Finalize dashboard development and project documentation.	• Developed multiple interactive dashboard pages with KPI cards, charts, and slicers • Created executive and supply chain analysis dashboards • Generated results and business insights from dashboard visualizations • Completed project documentation and dashboard explanation • Finalized Milestone 1 report, presentation, and dashboard outputs
Week3	• Develop the Customer Analysis Dashboard. • Create the Delivery Status Dashboard. • Analyze customer behavior and delivery performance. • Generate business insights from dashboards.	• Customer Analysis Dashboard completed. • Delivery Status Dashboard completed. • Customer behavior and delivery performance analyzed. • Business insights generated from dashboard visualizations

Week4	• Develop the Regional & Customer Insights Dashboard. • Review and optimize all dashboards. • Prepare the final report and presentation. • Verify dashboard accuracy and complete project submission.	• Regional & Customer Insights Dashboard completed. • All five dashboards reviewed and finalized. • Final report and presentation completed. • Project documentation verified and successfully submitted.
Week5	• Review the complete Power BI project. • Validate dashboard accuracy and consistency. • Finalize the project report and presentation. • Prepare for project demonstration and presentation. • Submit the final internship project.	• Reviewed all dashboards and verified the results. • Checked dashboard consistency, formatting, and accuracy. • Final report and presentation finalized successfully. • Prepared for the project presentation and viva. • Successfully completed and submitted the internship project.
Week6	• Review the complete project and final dashboard outputs. • Organize the final project files and supporting documents. • Upload and maintain the project files in GitHub and GitLab repositories. • Prepare the final project demonstration and explain the dashboard insights. • Incorporate final feedback and ensure all project deliverables are complete.	• Reviewed the complete Power BI project and all seven dashboards. • Organized the final dashboard files, report, presentation, and supporting project materials. • Prepared the GitHub repository and worked on the GitLab repository for project submission. • Prepared the final project demonstration and dashboard explanations. • Incorporated final feedback and verified that the project deliverables were ready for final submission.

5a. Key Milestones :

Milestone	Description	Date Achieved
Project Kickoff	Planned the project, defined the objectives, selected the DataCo Supply Chain dataset, and understood the project requirements.	Week 1
Prototype / First Draft	Completed data preprocessing, data cleaning, created initial KPIs, and developed the first Power BI dashboard prototype.	Week2

Mid-Term Review	Enhanced dashboards by adding interactive visualizations, reviewed dashboard performance, and incorporated improvements based on project progress.	Week3
Final Submission	Completed all five Power BI dashboards, finalized project documentation, prepared the report, and verified the dashboard outputs.	Week4
Presentation	Prepared the final presentation, demonstrated the project, explained business insights, and successfully completed the internship project submission.	Week5
Project Documentation & Repository Submission	Organized the completed project files and documentation, uploaded the project materials to the GitHub repository, worked on the GitLab repository, and prepared the final project materials for submission and demonstration.	Week6

Milestone 1 – Data Preprocessing and Dataset Preparation

In the first milestone, I focused mainly on **understanding, preprocessing, and cleaning the DataCo Supply Chain Dataset**. I first collected the dataset in CSV format and examined its structure, columns, data types, and the type of supply chain information available. I identified important fields related to orders, customers, products, sales, profit, discounts, shipping, delivery status, markets, regions, categories, and departments.

I then performed the required **data preprocessing using Power Query Editor**. I corrected the data types of different columns, particularly date, text, integer, and decimal fields. I removed unnecessary columns, handled duplicate records and errors, renamed columns for better readability, replaced inconsistent values, and filtered unwanted records. I also performed required transformations such as splitting customer information and creating useful derived columns including Year and Quantity Range. After completing these transformations, I reviewed the cleaned dataset and prepared it in an analysis-ready format for Power BI dashboard development.

Milestone 2 – Development of First Two Dashboards

In the second milestone, I started the **Power BI dashboard development** using the cleaned and transformed dataset. I developed the first two dashboards: **Supply Chain Performance Dashboard** and **Product Performance & Pricing Dashboard**.

The **Supply Chain Performance Dashboard** was developed to provide an overall view of supply chain operations. I created KPI cards to display important measures such as **Total Orders, Total Sales, Total Profit/Benefit, and Average Actual Shipping Days**. I also created

visualizations to analyze **sales by market, delivery status, and sales trends over years**. These visuals helped provide an overall understanding of business performance and delivery efficiency.

The **Product Performance & Pricing Dashboard** was developed to analyze product-related performance. I created visuals for **top-performing products, product categories, departments, product prices, and sales**. I also used a scatter chart to analyze the relationship between **product price and sales**. This milestone helped me understand how different products, categories, departments, and pricing levels contribute to overall business performance.

Milestone 3 – Development of Customer, Delivery and Regional Dashboards

In the third milestone, I expanded the project by developing three additional dashboards: **Customer & Geographic Analysis Dashboard, Delivery & Shipping Performance Dashboard, and Regional & Market Performance Dashboard**.

The **Customer & Geographic Analysis Dashboard** focused on understanding customer distribution and customer-related sales performance. I analyzed **customer locations, states, cities, customer segments, sales per customer, and product contribution**. Maps, charts, KPI cards, and other visuals were used to understand geographic customer patterns and customer performance.

The **Delivery & Shipping Performance Dashboard** focused on analyzing **delivery status, late deliveries, scheduled shipping days, shipping modes, market-wise profitability, and yearly shipping performance**. I created appropriate visualizations to compare different shipping modes and identify delivery-related performance patterns. This helped highlight areas where logistics and delivery efficiency could be improved.

The **Regional & Market Performance Dashboard** focused on comparing the performance of different **markets, customer segments, and shipping modes**. I created KPI cards for important business measures, a gauge chart for comparing sales with the defined sales target, and other visuals for market-wise sales and customer segment performance. I also analyzed order distribution across shipping modes. By completing these three dashboards, I was able to analyze the project from **customer, delivery, geographic, regional, and market perspectives**.

Milestone 4 – Development of Order, Discount and Profitability Dashboards

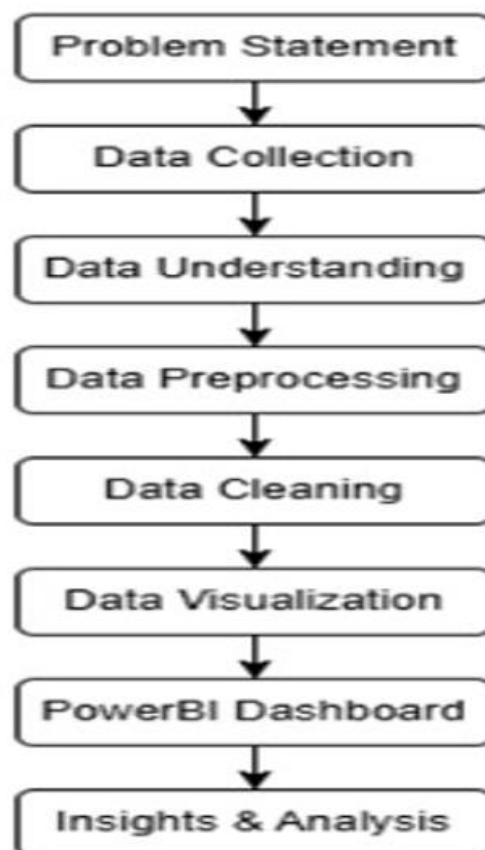
In the fourth milestone, I completed the final two dashboards: **Order & Discount Performance Dashboard** and **Profitability & Order Insights Dashboard**. These dashboards were developed to complete the remaining areas of supply chain analysis and provide a detailed view of order behavior, discount patterns, and profitability.

The **Order & Discount Performance Dashboard** focused on **order quantity, quantity ranges, order status, discount rates, sales, markets, and shipping modes**. I created a funnel chart to analyze **Orders by Quantity Level** and used other visualizations to study order-status distribution and the relationship between discount rates and sales. A decomposition analysis was also used to break down sales across different business dimensions.

The **Profitability & Order Insights Dashboard** focused on **profit ratio, total discounts, shipping-mode profitability, departmental sales, discount trends, and order status**. I created KPI cards and different charts to analyze profitability and performance trends over years. After completing both dashboards, I reviewed all the visuals and ensured that the seven dashboards together covered different and non-repetitive aspects of the supply chain dataset.

By the end of this milestone, I successfully completed the **full set of seven interactive Power BI dashboards** and generated business insights from the analysis. The completed dashboards provided a comprehensive view of **supply chain performance, products, pricing, customers, geography, delivery, shipping, markets, orders, discounts, and profitability**, forming the final analytical output of the project.

5b. Project Execution Details



Step 1: Problem Statement

The project was initiated to improve **supply chain visibility and business performance analysis** using Microsoft Power BI. The main objective was to transform raw supply chain data into meaningful and interactive visual insights that could help understand different areas of business operations. The analysis focused on **sales, orders, products, customers, pricing, discounts, profit, markets, shipping performance, delivery status, and regional performance**.

The project was designed to provide different business perspectives through multiple dashboards rather than combining all information into a single report. This approach makes it easier for management and users to monitor specific areas, identify performance gaps, compare business trends, and support **data-driven decision-making and operational improvement**.

Step 2: Data Collection

The **DataCo Supply Chain Dataset** was collected in CSV format from Kaggle. The dataset contains a large number of records covering different aspects of supply chain operations, including **customer information, product details, order transactions, sales, discounts, profit, shipping, delivery status, markets, regions, categories, and departments**.

The collected dataset was initially examined to understand the available fields and their relevance to the project requirements. The CSV dataset was then imported into **Microsoft Power BI**, where Power Query Editor was used for preprocessing, transformation, and data cleaning before developing the dashboards.

Step 3: Data Understanding

The dataset was carefully studied before beginning the visualization process. The available columns were examined based on their **business meaning, data type, relationship with other fields, and usefulness for dashboard analysis**.

Important fields related to **Order ID, Customer ID, Customer Full Name, Customer Segment, Customer City, Customer State, Product Name, Product Price, Category Name, Department Name, Sales, Order Item Quantity, Discount, Profit Ratio, Benefit per Order, Market, Order Region, Shipping Mode, Delivery Status, Late Delivery Risk, Order Status, Order Date, and Shipping Date** were identified.

The data was divided conceptually into areas such as **order analysis, product analysis, customer analysis, delivery and shipping analysis, market and regional analysis, discount analysis, and profitability analysis**. This understanding helped in selecting appropriate columns and visualizations for the seven dashboards.

Step 4: Data Preprocessing

Data preprocessing was performed using **Power Query Editor** to convert the raw dataset into a structured and analysis-ready format. Multiple transformation steps were applied to improve the consistency, readability, and usability of the dataset.

Major preprocessing activities included:

- **Changed Data Types:** Corrected the data types of date, text, integer, decimal, and other numerical columns according to their actual values.
- **Changed Type with Locale:** Used appropriate locale settings for date/time fields to correctly interpret date values.
- **Removed Unnecessary Columns:** Removed columns that were not required for the final seven dashboards and analysis.
- **Removed Duplicate Records:** Checked and removed duplicate records wherever applicable to improve data consistency.
- **Merged Columns:** Combined relevant fields where required to create meaningful information for analysis.
- **Renamed Columns:** Changed column names into clearer and more readable names for easier identification in Power BI.
- **Replaced Values:** Corrected or standardized inconsistent values using the Replace Values transformation.
- **Removed Errors:** Identified and removed error records generated during data transformation.
- **Filtered Rows:** Applied row-level filters wherever invalid or unnecessary records were identified.
- **Split Customer Name:** The Customer Full Name field was transformed into separate customer name components for better customer-level analysis.
- **Added Custom Columns:** Created additional columns required for analysis and dashboard development.
- **Inserted Year:** Extracted the year from the order date to support year-wise sales and performance analysis.
- **Added Conditional Column:** Created the **Quantity Range** classification to group orders according to their order quantity.
- **Sorted Rows:** Sorted relevant data to improve organization and analysis.
- **Final Column Review:** After dashboard development, the dataset was reviewed again and unnecessary columns were removed to maintain a cleaner dataset.

These preprocessing steps ensured that the data was **clean, consistent, readable, and suitable for creating accurate Power BI visualizations**.

Step 5: Data Cleaning

Data cleaning was performed after and along with preprocessing to improve the overall quality of the dataset. The objective was to reduce data quality problems that could affect calculations, KPIs, charts, and dashboard insights.

The following activities were carried out:

- Removed unnecessary and unused columns.
- Removed duplicate records where identified.
- Handled blank and missing values.
- Removed records containing transformation errors.
- Corrected incorrect or inconsistent data types.
- Standardized column names for better readability.
- Replaced inconsistent values where required.
- Filtered invalid or unwanted records.
- Checked numerical and date fields for proper formatting.
- Verified important sales, order, customer, product, and shipping fields.
- Reviewed the cleaned dataset before connecting it to the dashboard visuals.
- Performed a final validation to ensure that the dashboard calculations were based on properly prepared data.

After cleaning, the transformed dataset was connected to Power BI for analysis and visualization.

Step 6: Data Visualization

After preprocessing and cleaning, different Power BI visualizations were selected according to the type of business information being analyzed. The objective was to make complex supply chain information easier to understand and compare.

The following visualizations were used across the seven dashboards:

- **KPI Cards** – Used to display important numerical indicators such as total sales, total orders, profit/benefit, quantity, and average values.
- **Bar Charts** – Used to compare products, categories, departments, markets, states, and other business dimensions.
- **Line Charts** – Used to analyze yearly trends such as sales, shipping performance, benefits, and discounts.
- **Pie Charts** – Used to show the distribution of orders across shipping modes.
- **Donut Charts** – Used to analyze delivery status, customer segments, and order-status distribution.
- **Treemaps** – Used to show product, department, and profitability contributions.
- **Scatter Charts** – Used to analyze relationships such as product price vs sales and discount rate vs sales.

- **Maps/Filled Maps** – Used for geographical customer and state-level analysis.
- **Tables** – Used to provide detailed comparisons of markets and customer segments.
- **Waterfall Chart** – Used to analyze market-wise contribution to overall benefit.
- **Decomposition Tree** – Used to break down sales by market, category, and shipping mode.
- **Funnel Chart** – Used to analyze orders according to quantity levels.
- **Gauge Chart** – Used to compare actual sales against the defined sales target.
- **Slicers** – Used to allow interactive filtering, particularly for year-based analysis.

Step 7: Dashboard Development

After completing the data preparation and visualization process, **seven interactive Power BI dashboards** were developed. Each dashboard was designed for a different business perspective and contains relevant KPIs and visualizations.

Dashboard 1 – Supply Chain Performance Dashboard

This dashboard provides an overall view of supply chain performance.

- Displays **Total Orders, Total Benefit/Profit, Total Sales, and Average Actual Shipping Days** through KPI cards.
- Analyzes overall **Delivery Status** using a donut chart.
- Compares **Sales by Market** to identify high-performing markets.
- Displays **Sales Trend Over Time** to analyze yearly changes in sales.
- Helps management understand overall business performance and delivery efficiency.

Dashboard 2 – Product Performance & Pricing Dashboard

This dashboard focuses on product-level sales, pricing, categories, and departments.

- Displays the **Total Products** and **Average Product Price** through KPI cards.
- Identifies **Top 10 Products by Sales**.
- Compares **Sales by Product Category**.
- Analyzes **Sales by Department** using a treemap.
- Uses a **Product Price vs Sales Scatter Chart** to understand the relationship between product pricing and sales.
- Helps identify high-performing products and supports product pricing and inventory-related decisions.

Dashboard 3 – Customer & Geographic Analysis Dashboard

This dashboard analyzes customer distribution and customer-related sales performance.

- Displays **Total States** and **Average Sales per Customer** through KPI cards.
- Uses a geographical map to show **Customer Distribution by State**.
- Compares **Sales by Customer City**.
- Identifies the **Top 10 States by Sales**.
- Analyzes **Customer Segment Distribution** using a donut chart.
- Uses a treemap to analyze **Profit Contribution by Product**.
- Helps understand customer concentration, customer segments, and geographical sales patterns.

Dashboard 4 – Delivery & Shipping Performance Dashboard

This dashboard focuses on delivery efficiency and shipping performance.

- Displays **Total Late Deliveries** and **Average Scheduled Shipping Days**.
- Provides a **Year Slicer** for year-wise analysis.
- Compares **Delivery Status by Shipping Mode** using a 100% stacked chart.
- Analyzes **Profit/Benefit Contribution by Market** using a waterfall chart.
- Displays **Shipping Performance Trend by Year**.
- Helps identify delivery issues, compare shipping modes, and understand market-wise logistics performance.

Dashboard 5 – Regional & Market Performance Dashboard

This dashboard focuses on regional, market, customer segment, and shipping performance.

- Displays **Average Benefit per Order** and **Total Markets** using KPI cards.
- Uses a **Gauge Chart** to compare actual sales with the predefined sales target.
- Compares **Average Sales by Market**.
- Provides a detailed **Customer Segment Performance** table containing sales, benefit, and order information.
- Provides a **Market-wise Sales Summary** for comparing different markets.
- Uses a pie chart to analyze **Orders by Shipping Mode**.
- Helps identify strong markets, customer segments, sales achievement, and order distribution.

Dashboard 6 – Order & Discount Performance Dashboard

This dashboard focuses on order quantity, discount behavior, and sales analysis.

- Displays **Total Quantity Sold** and **Average/Discount Rate information** through KPI cards.

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- Uses a **Funnel Chart** to analyze **Orders by Quantity Level**.
- Compares **Order Status Distribution by Market**.
- Uses a scatter chart to analyze the relationship between **Discount Rate and Sales by Product**.
- Uses a **Decomposition Tree** to break down sales through different dimensions such as market, category, and shipping mode.
- Helps understand customer order quantity patterns, discount behavior, and factors contributing to sales.

Dashboard 7 – Profitability & Order Insights Dashboard

This dashboard focuses on profitability, discounts, shipping modes, departments, and order status.

- Displays **Average Profit Ratio** and **Total Discount Amount** through KPI cards.
- Compares **Average Benefit by Shipping Mode across Years**.
- Analyzes **Sales by Department over Years**.
- Displays the **Discount Trend over Time**.
- Uses a donut chart to analyze **overall Order Status Distribution**.
- Helps understand profitability, discount impact, departmental sales trends, and the overall order lifecycle.

Step 8: Insights & Business Recommendations

After completing the seven dashboards, the analyzed data provided several important business insights.

Key Insights

- More than **180K orders** were processed, indicating a large volume of supply chain activity.
- **Europe** emerged as one of the strongest markets in terms of sales performance.
- **Standard Class** accounted for the largest share of orders among shipping modes.
- Late deliveries represent an important area for logistics improvement.
- The **Consumer customer segment** contributed a major share of customer activity.
- Product and category analysis helped identify important sales contributors.
- Product pricing and sales showed different patterns across products.
- Sales and performance varied across different markets, states, cities, categories, and departments.
- Discount analysis helped identify the relationship between discount levels and product sales.

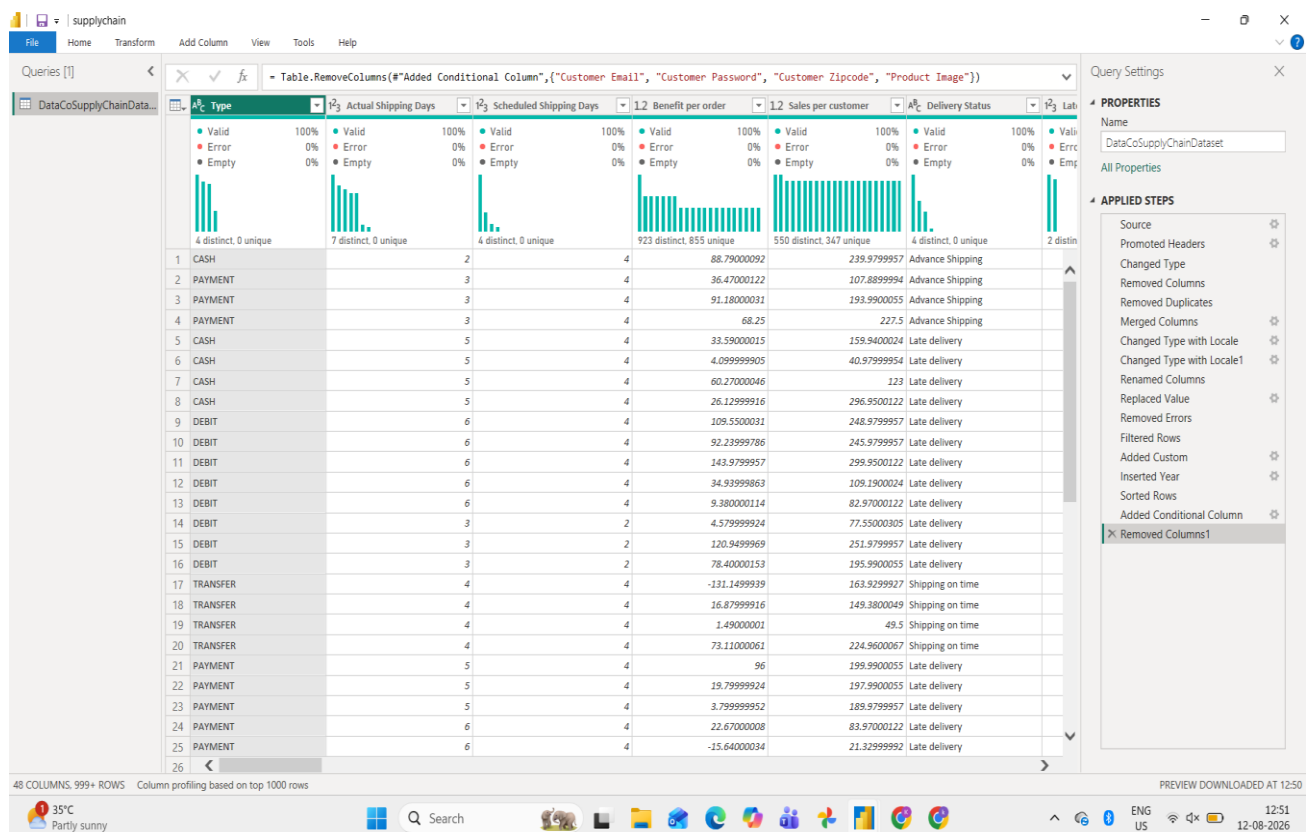
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- Year-wise analysis helped identify changes in sales, shipping performance, department performance, and discount trends.
- Profitability analysis helped compare the contribution of different products, markets, departments, and shipping modes.

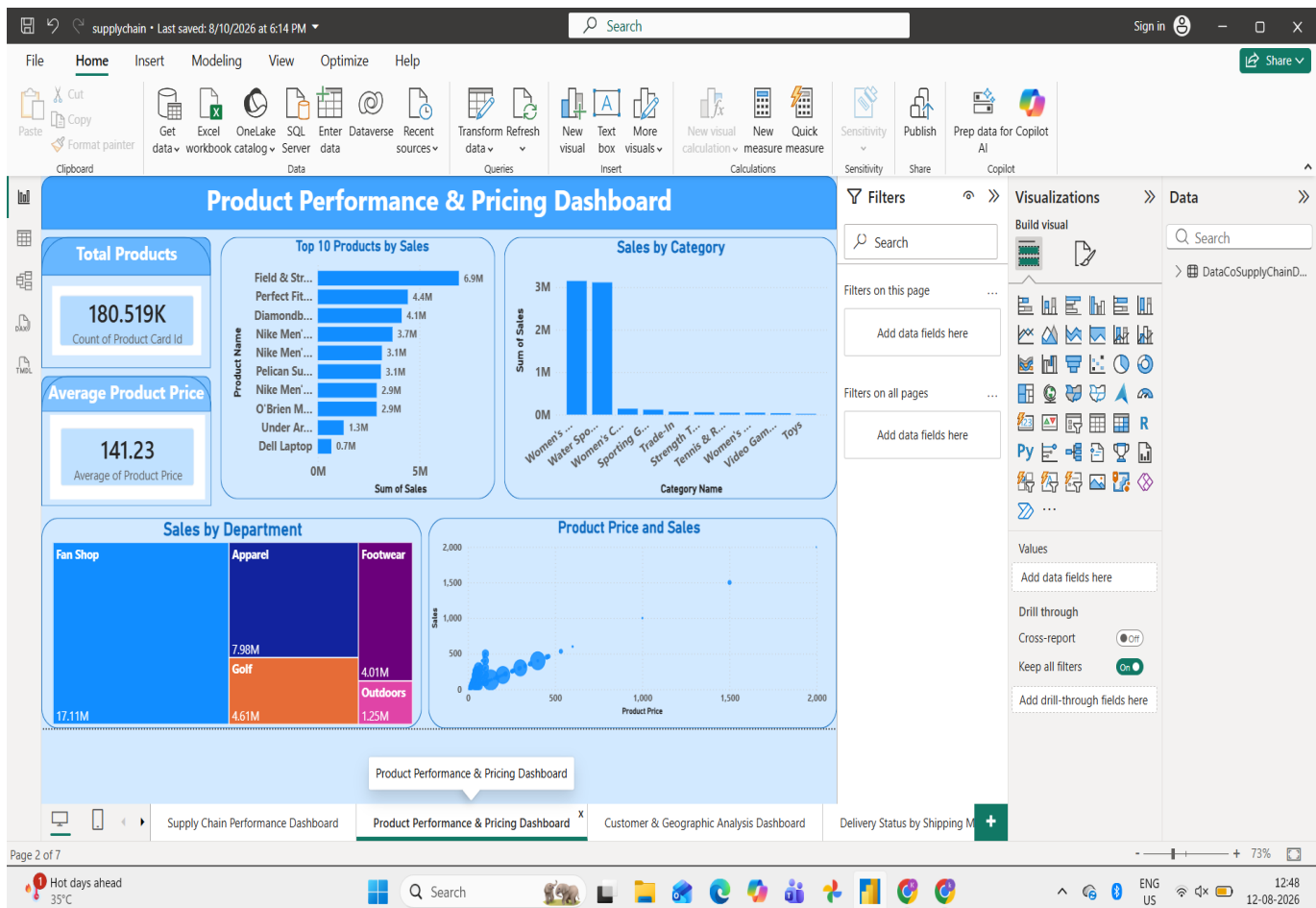
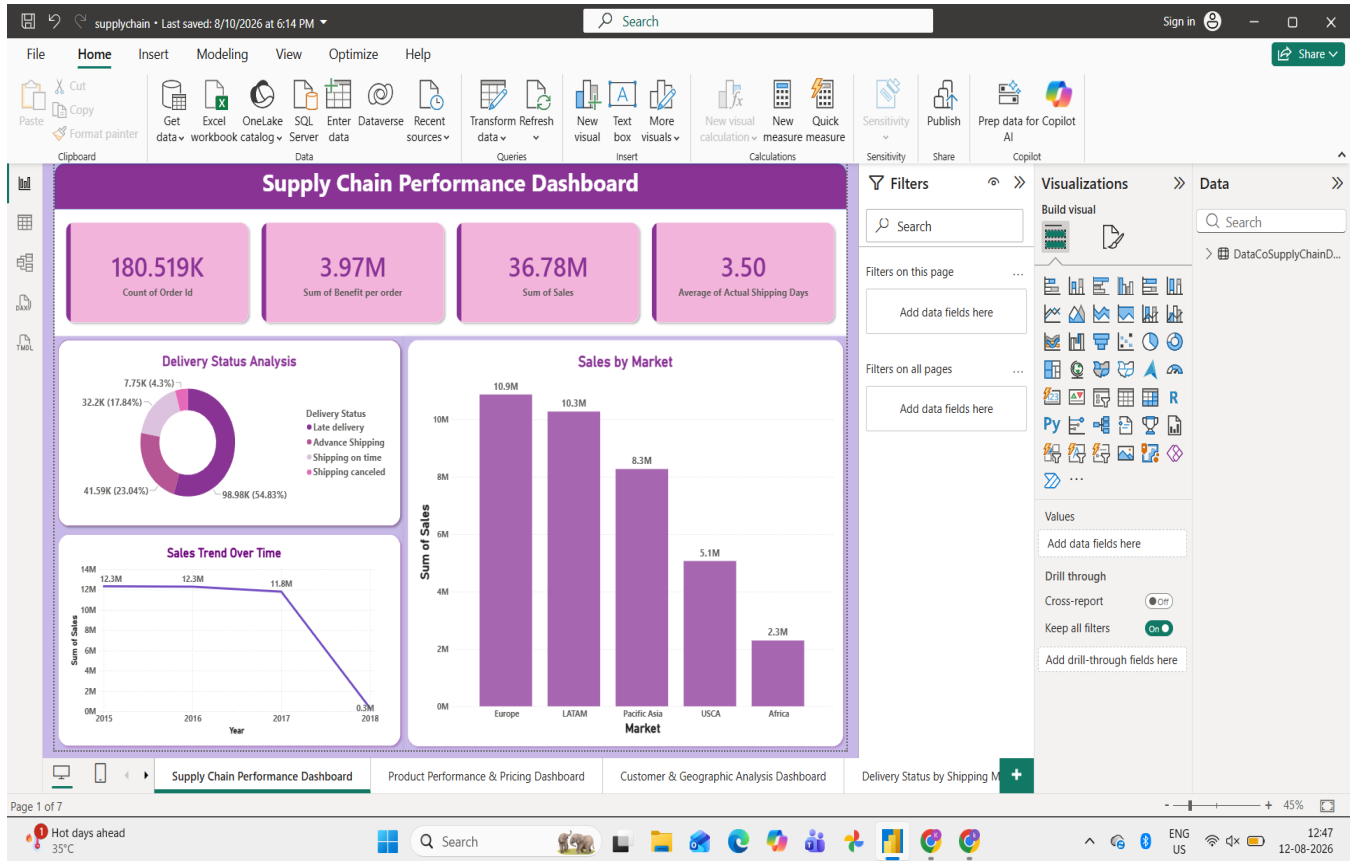
Business Recommendations

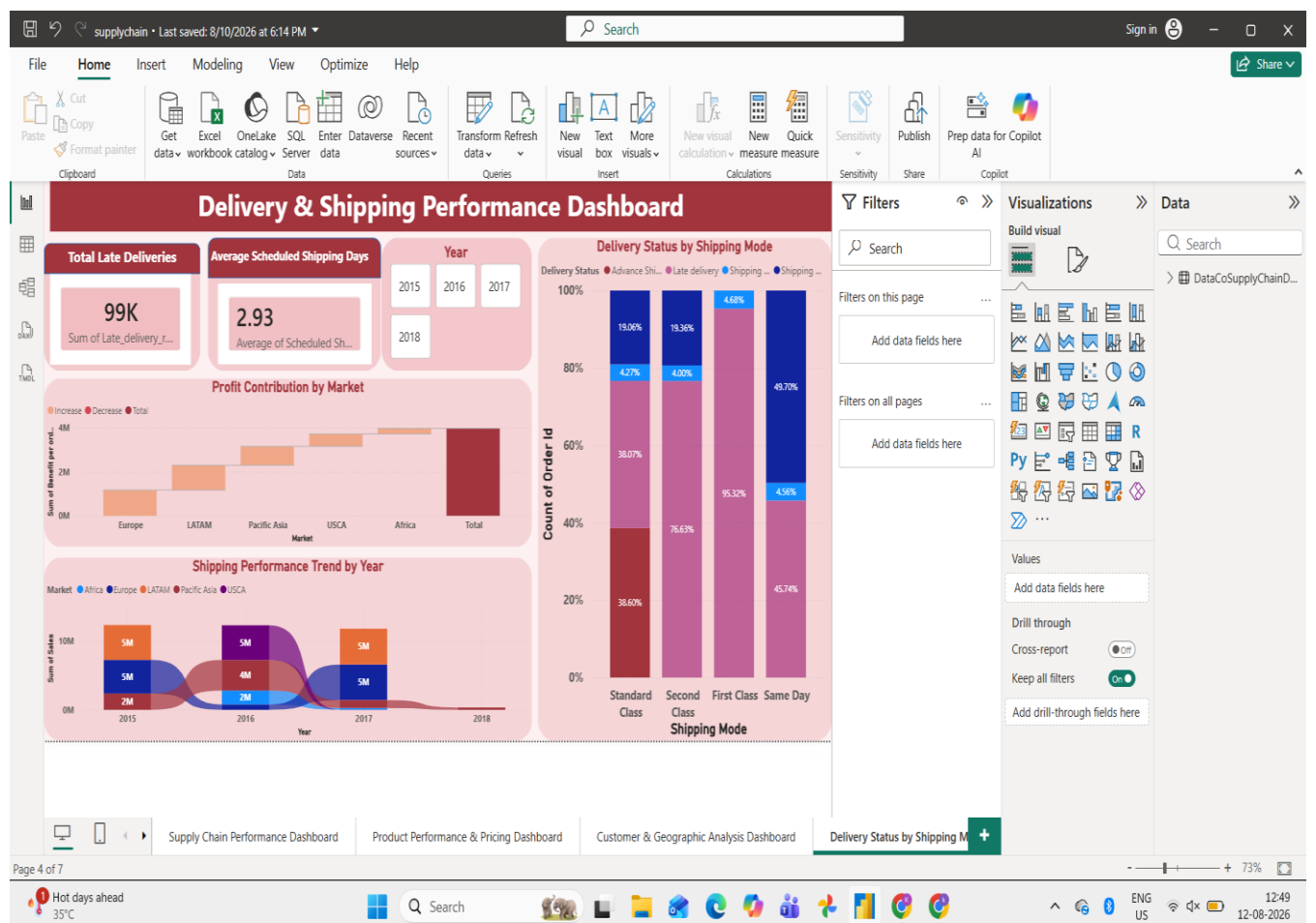
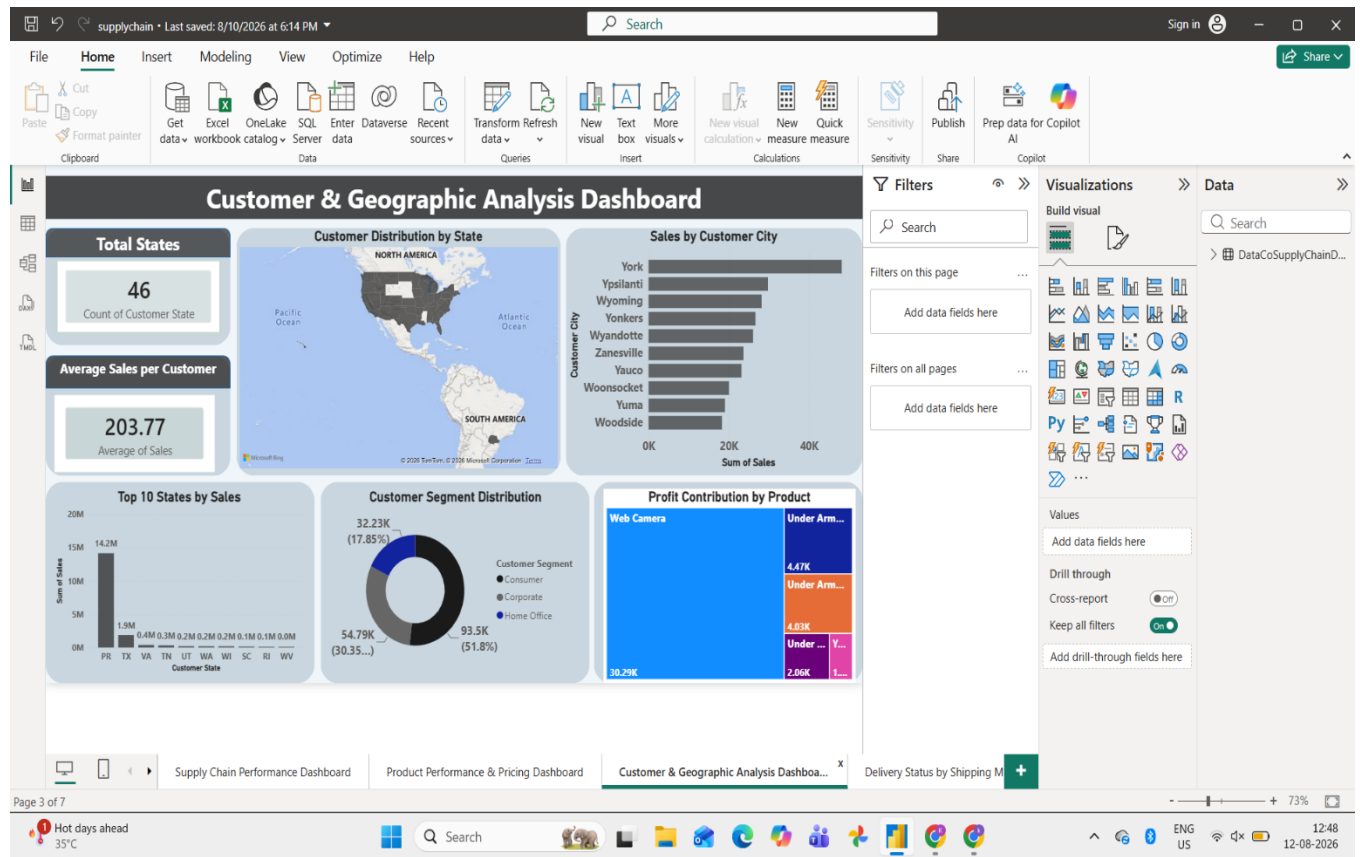
- Improve logistics and delivery planning to reduce late deliveries.
- Monitor shipping modes with higher delivery issues and improve their operational efficiency.
- Maintain sufficient inventory for high-performing products and categories.
- Focus marketing and sales strategies on high-performing markets and customer segments.
- Regularly monitor low-performing markets, products, and departments.
- Review discount strategies to ensure that increased discounts contribute effectively to sales.
- Use product pricing and sales analysis to support better pricing decisions.
- Monitor profitability across shipping modes, products, and markets.
- Use the interactive Power BI dashboards for continuous performance monitoring and **data-driven strategic decision-making**.

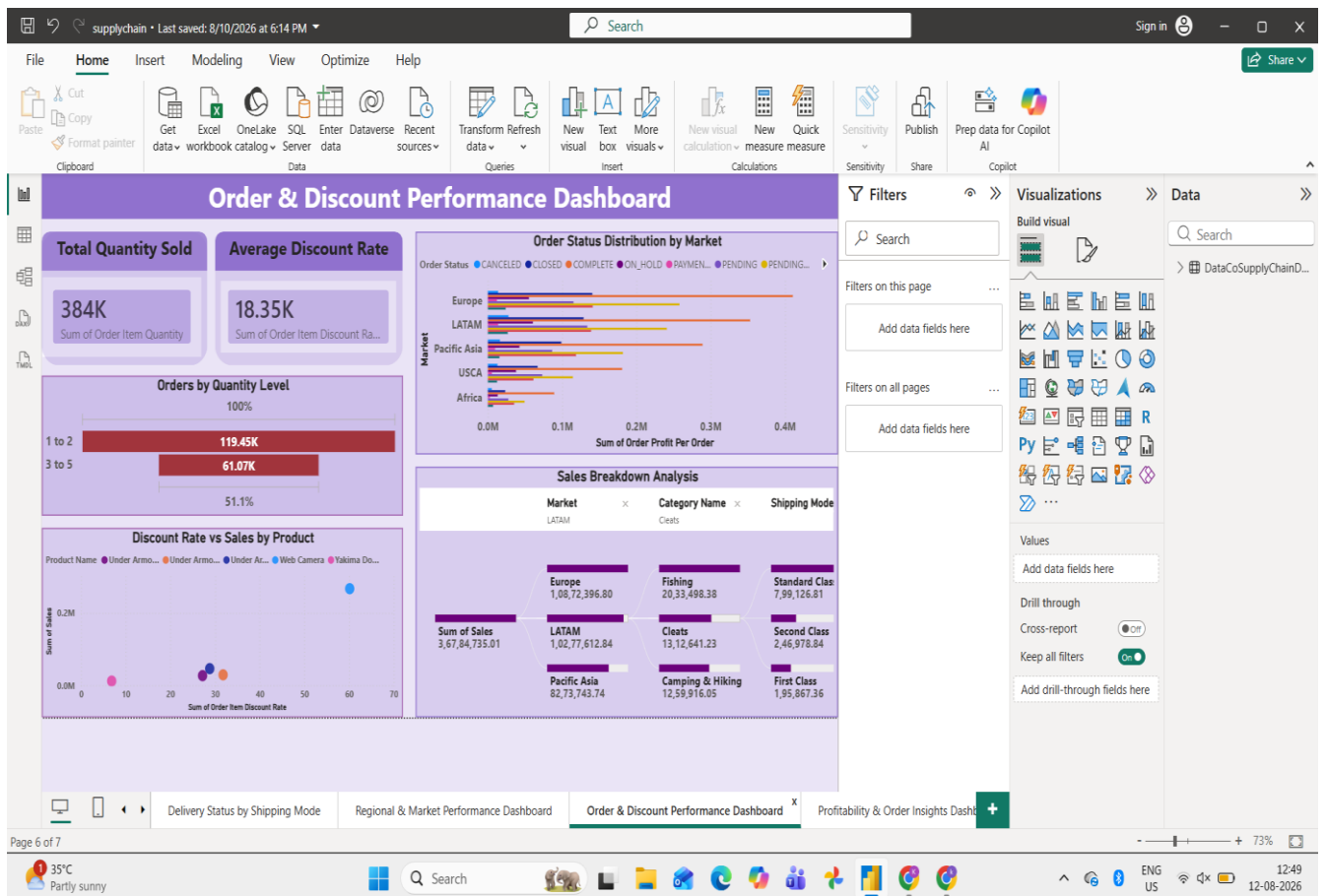
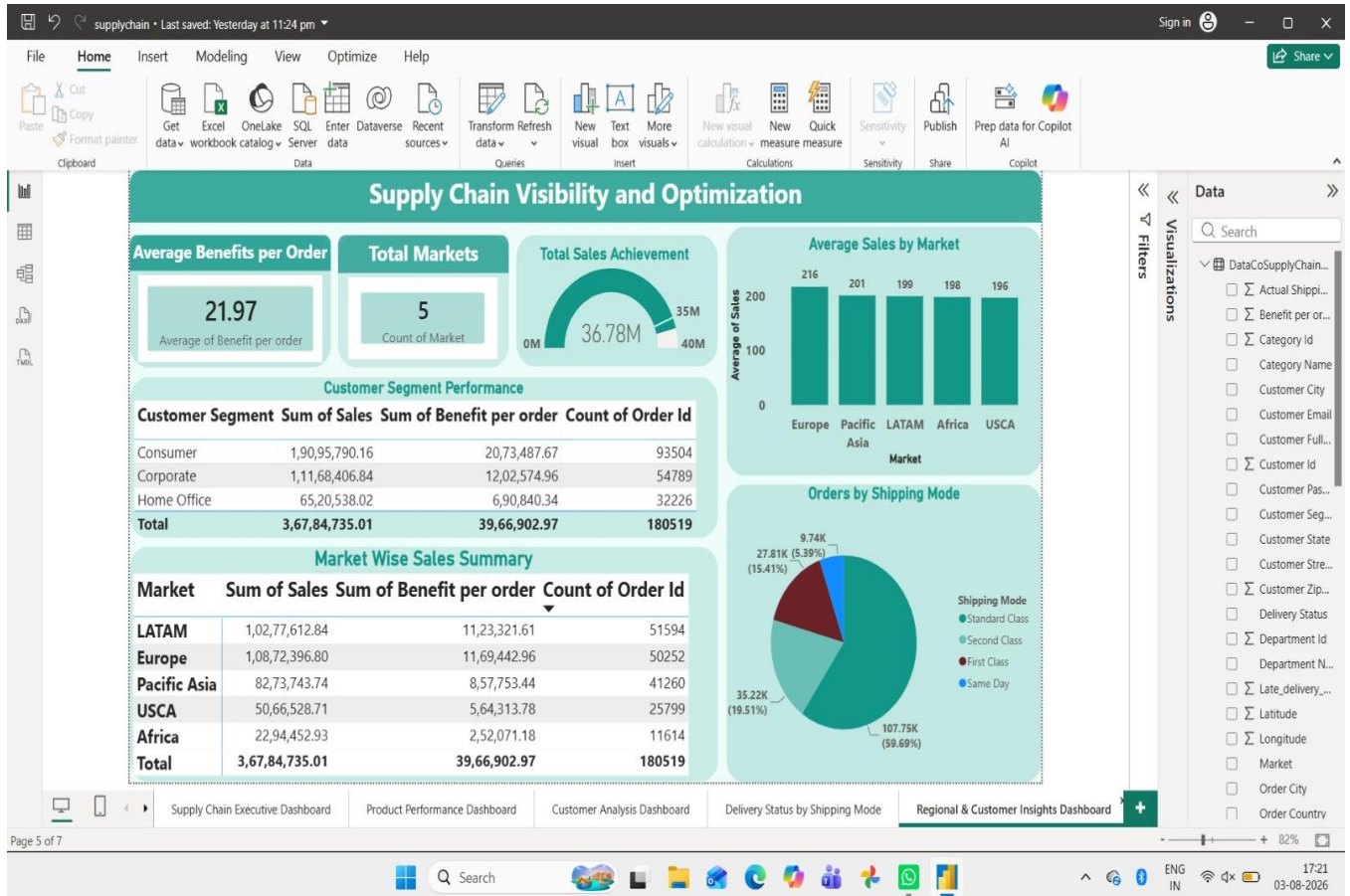
6. Snapshots/Screenshots :

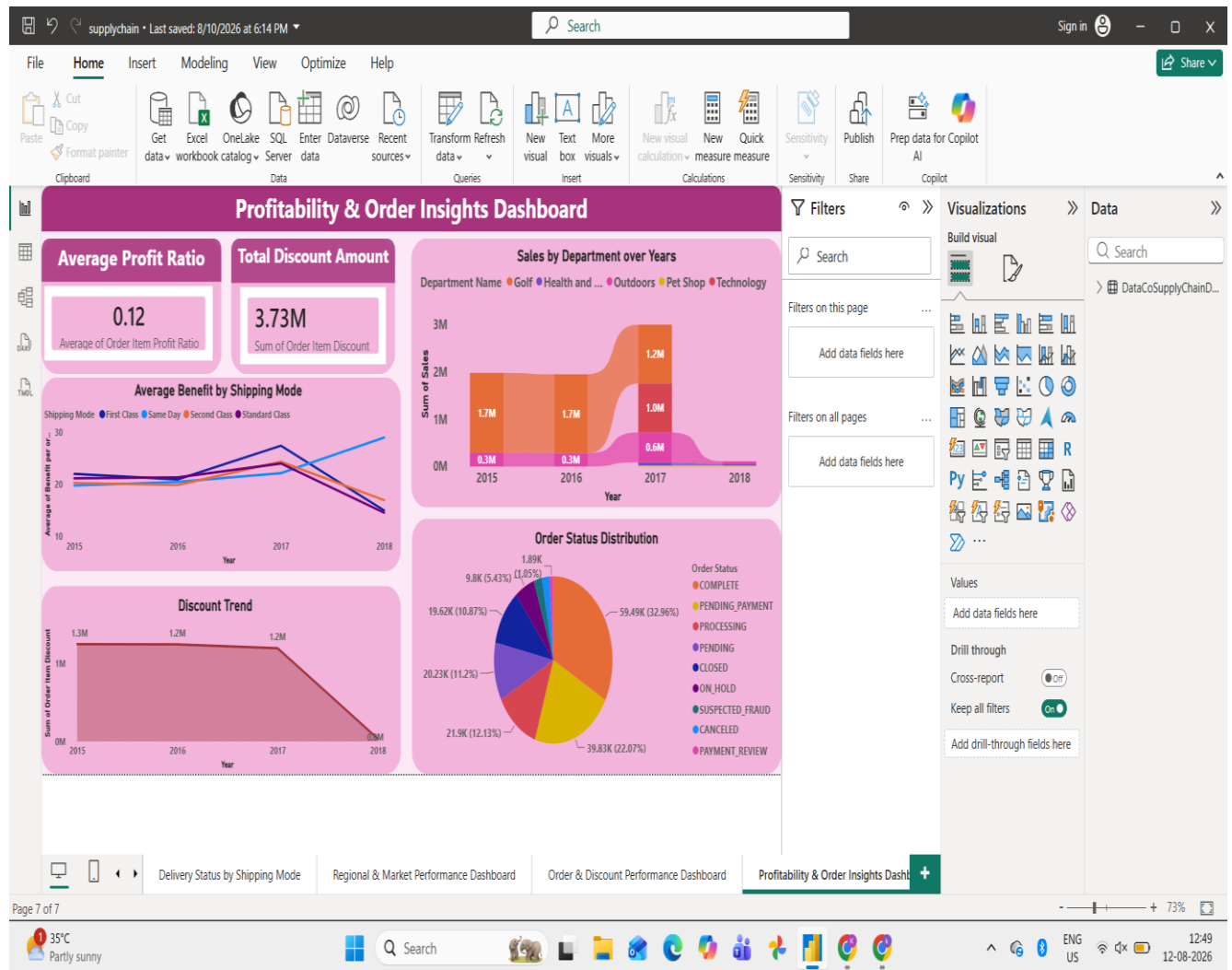


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7. Challenges Faced :

1. Understanding the Dataset :

One of the major challenges was understanding the large DataCo Supply Chain Dataset, which contains numerous attributes related to customers, products, orders, shipping, sales, and markets. Identifying the relevant columns and understanding the relationships between different data fields required careful analysis before starting the dashboard development process.

2. Data Preprocessing and Cleaning :

The dataset contained missing values, duplicate records, inconsistent data types, and unnecessary columns. Cleaning and transforming the dataset using **Power Query Editor** required significant effort to ensure data accuracy and consistency. Proper preprocessing was essential for generating reliable dashboard results.

3. Selecting Appropriate Visualizations :

Choosing the right Power BI visualizations for different business metrics was another challenge. Different chart types such as KPI cards, bar charts, line charts, donut charts, treemaps, scatter plots, maps, and waterfall charts were carefully selected to present information clearly and effectively according to business requirements.

4. Designing Professional and Interactive Dashboards :

Developing visually appealing and user-friendly dashboards required careful planning of colors, layout, spacing, alignment, and formatting. It was also important to avoid repeating the same visualizations across different dashboards while maintaining a consistent professional design throughout the project.

5. Generating Business Insights :

Analyzing the dashboard results and extracting meaningful business insights was one of the most important challenges. Sales trends, customer behavior, product performance, shipping efficiency, and regional business performance were carefully interpreted to provide valuable recommendations that support better business decision-making.

8. Learnings & Skills Acquired :

8.1 Technical Skills :

1. Microsoft Power BI Dashboard Development :

Gained practical experience in using Microsoft Power BI to design and develop interactive business intelligence dashboards. Learned how to create professional reports using various visualizations, KPI cards, filters, slicers, and dashboard navigation to present business information effectively.

2. Data Cleaning and Transformation :

Developed hands-on experience in Power Query Editor for performing data preprocessing tasks such as correcting data types, handling missing values, removing duplicate records, filtering invalid data, renaming columns, splitting customer names, and transforming raw data into an analysis-ready format.

3. Interactive Data Visualization :

Improved technical skills in creating different Power BI visualizations, including KPI Cards, Bar Charts, Line Charts, Pie Charts, Donut Charts, Treemaps, Scatter Charts, Filled Maps, Waterfall Charts, Ribbon Charts, Tables, Matrix, Gauge Charts, and Slicers. Learned how to select appropriate visualizations based on different business requirements.

4. Business Reporting and Dashboard Design :

Enhanced knowledge of designing professional, user-friendly, and interactive dashboards by applying proper layouts, color themes, formatting, alignment, and visual consistency.

Learned to present complex business data in a simple and understandable format for management and stakeholders.

5. Business Insight Generation :

Developed the ability to analyze supply chain data and generate meaningful business insights related to sales performance, customer behavior, product performance, delivery status, shipping efficiency, and regional performance. These insights support data-driven decision-making and improve overall business operations.

6. Analytical and Problem-Solving Skills :

Strengthened analytical thinking and problem-solving abilities by identifying data quality issues, selecting appropriate visualization techniques, interpreting dashboard results, and providing business recommendations based on data analysis.

8.2 Analytical and Problem-Solving Skills :**1. Data Analysis :**

Developed the ability to analyze large DataCo Supply Chain datasets and identify meaningful business trends related to sales, profit, customer behavior, product performance, shipping efficiency, and regional business performance. Learned to interpret complex business data and convert it into valuable insights.

2. Data Quality Management :

Strengthened problem-solving skills by identifying and resolving data quality issues such as missing values, duplicate records, inconsistent data types, and invalid data. Applied appropriate data cleaning and preprocessing techniques using Power Query Editor to ensure accurate and reliable analysis.

3. Visualization and Analytical Thinking :

Improved analytical thinking by selecting the most appropriate Power BI visualizations for different business requirements. Evaluated various chart types to ensure that business information was presented in a clear, meaningful, and easy-to-understand format.

4. Business Decision Support :

Developed the ability to interpret dashboard results and generate meaningful business insights. Analyzed sales trends, customer segments, delivery performance, product performance, and regional sales to support effective business planning and data-driven decision-making.

5. Problem-Solving Approach :

Enhanced problem-solving skills by addressing real-world business challenges during dashboard development. Successfully identified operational issues, optimized dashboard design, improved report usability, and provided practical recommendations based on analytical findings.

6. Critical Thinking and Continuous Improvement :

Strengthened critical thinking by continuously reviewing dashboard outputs, validating results, and refining visualizations to improve accuracy and presentation quality. This process helped in delivering professional dashboards with reliable business insights.

8.3 Future Enhancements :

1. Real-Time Data Integration :

In the future, the dashboard can be enhanced by integrating real-time supply chain data from enterprise systems such as ERP or cloud databases. This will enable live monitoring of sales, inventory, orders, and delivery performance, allowing organizations to make faster and more informed business decisions.

2. Machine Learning-Based Demand Forecasting :

Advanced machine learning algorithms can be incorporated to predict future product demand, sales trends, and inventory requirements. This will help organizations optimize inventory planning, reduce stock shortages, minimize excess inventory, and improve overall supply chain efficiency.

3. Advanced Dashboard Features :

The dashboards can be further enhanced by adding advanced Power BI features such as drill-through reports, bookmarks, dynamic tooltips, advanced filters, and interactive navigation. These features will allow users to perform detailed analysis and explore business data more effectively.

4. Automated Data Refresh and Reporting :

Future improvements can include automated data refresh and scheduled report generation. This will ensure that dashboards always display the latest business information without requiring manual updates, thereby improving reporting efficiency and reducing operational effort.

5. Enhanced Business Analysis :

The dashboard can be expanded by including additional analytical modules such as inventory management, supplier performance analysis, warehouse operations, logistics optimization, and customer satisfaction analysis. These enhancements will provide a more comprehensive view of supply chain operations and support strategic business planning.

6. Scalability and Business Intelligence :

Future versions of the project can be designed to support larger datasets, multiple business locations, and cloud-based deployment. This will improve scalability, enable organization-wide reporting, and provide a centralized Business Intelligence platform for continuous performance monitoring and decision-making.

8.4 Soft Skills and Team Collaboration :**1. Communication and Presentation Skills :**

Throughout the internship, communication and presentation skills were significantly improved by preparing project documentation, creating professional presentations, and explaining dashboard results clearly. This experience enhanced the ability to present technical information in a simple and understandable manner to different audiences.

2. Time Management :

Strong time management skills were developed by completing each project milestone according to the planned schedule. Tasks such as data preprocessing, dashboard development, documentation, and presentation preparation were effectively organized and completed within the given deadlines.

3. Problem-Solving and Critical Thinking :

The project provided opportunities to solve real-world business and technical challenges related to data quality, dashboard design, and business analysis. These activities strengthened analytical thinking, critical thinking, and the ability to identify practical solutions based on data.

4. Documentation and Project Organization :

Improved the ability to prepare well-structured project reports, maintain organized documentation, and present project progress in a professional manner. Proper documentation ensured that every stage of the project was clearly recorded and easy to understand.

5. Collaboration and Project Planning :

Enhanced collaboration and project planning skills by following a structured development approach throughout the internship. Regular review of project milestones, continuous improvements, and systematic planning helped in successfully completing the project and achieving the internship objectives.

6. Professional Development :

The internship helped build confidence, responsibility, adaptability, and a professional work approach. It also improved the ability to manage tasks independently, learn new technologies quickly, and apply analytical skills to solve business problems effectively.

8.5 Domain Knowledge and Application :

1. Supply Chain Management Knowledge :

The internship provided practical exposure to Supply Chain Management concepts, including sales analysis, order management, customer analysis, product performance, shipping operations, and delivery status monitoring. This helped in understanding how different supply chain activities are connected and contribute to overall business performance.

2. Business Performance Analysis :

Developed the ability to analyze key business metrics such as sales, profit, customer behavior, product categories, market performance, and delivery efficiency. Learned how these metrics can be used to evaluate organizational performance and identify areas for improvement.

3. Application of Business Intelligence :

Applied Business Intelligence (BI) techniques using Microsoft Power BI to transform raw supply chain data into meaningful visual reports. Interactive dashboards were created to simplify complex business information and support effective business analysis.

4. Decision-Making and Business Insights :

Gained an understanding of how dashboard visualizations support both operational and strategic decision-making. The generated insights help management monitor business performance, identify trends, improve supply chain efficiency, and make informed business decisions.

5. Real-World Business Application :

Applied theoretical concepts to a real-world supply chain dataset by performing data preprocessing, visualization, dashboard development, and business analysis. This practical experience improved the ability to solve business problems using data-driven approaches and Business Intelligence tools.

6. Team Collaboration :

Worked collaboratively throughout the internship by following project milestones, organizing project tasks, reviewing progress, and contributing to documentation, dashboard development, and presentation activities. This experience strengthened teamwork, coordination, and professional collaboration skills.

9. Testimonials from team :

Team Member : Kapu Tejaswini

I independently worked on the complete **Supply Chain Visibility and Optimization** project. I collected the **DataCo Supply Chain Dataset** and performed data preprocessing and cleaning using **Power Query Editor**. I corrected data types, removed unnecessary columns, handled duplicate and error records, transformed the data, renamed columns, replaced inconsistent values, filtered records, and created useful derived columns such as **Year and Quantity Range**.

I developed and finalized all **seven Power BI dashboards: Supply Chain Performance Dashboard, Product Performance & Pricing Dashboard, Customer & Geographic Analysis Dashboard, Delivery & Shipping Performance Dashboard, Regional & Market Performance Dashboard, Order & Discount Performance Dashboard, and Profitability & Order Insights Dashboard**. I created KPIs, charts, maps, tables, and interactive visualizations, analyzed the dashboard results, generated business insights and recommendations, and prepared the **project report, documentation, presentation, and final project materials**. Through this project, I gained practical experience in **data preprocessing, Power BI visualization, business analysis, and data-driven decision-making**.

10.Conclusion :

The **DataCo Supply Chain Visibility and Optimization** project successfully transformed a large raw supply chain dataset into meaningful and interactive business insights using **Power BI**. The project involved systematic **data collection, understanding, preprocessing, cleaning, transformation, visualization, and analysis** using Power Query Editor and Power BI. A total of **seven interactive dashboards** were developed to analyze different business perspectives, including supply chain performance, product and pricing performance, customer and geographic analysis, delivery and shipping performance, regional and market performance, order and discount performance, and profitability and order insights.

The analysis provided valuable insights into **sales, orders, products, customers, markets, shipping, delivery, discounts, and profitability**. The company processed more than **180K orders**, generated approximately **36.78 Million in sales**, and achieved around **3.97 Million in total benefit/profit**. Europe was identified as a strong-performing market, while the Consumer segment represented a major share of customer activity. The analysis also identified **Standard Class as the most frequently used shipping mode**, while the high number of late deliveries highlighted the need for improvements in logistics and delivery planning. Product, department, regional, customer, and discount analysis further helped identify important performance patterns.

Overall, the seven dashboards provide a **comprehensive and interactive reporting solution** for understanding supply chain performance from multiple perspectives. The project demonstrates how **Business Intelligence and data visualization** can convert raw data into actionable insights, helping organizations monitor KPIs, identify trends and performance gaps, improve operational efficiency, optimize business strategies, and support **faster and more effective data-driven decision-making**.

11. Acknowledgements :

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